

A Preliminary Taxonomy and Evaluation Methodology for Recursive Bioadaptive Media

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Abstract

This paper examines recursive bioadaptive media as a class of perception-modulating systems that use psychophysiological and affective signals (e.g., EEG, HRV, GSR, eye tracking) to alter media parameters in real time. Unlike augmenting human sensation through prosthetic extension, these systems operate through recursive bioadaptive loops that modulate cognition, affect, and embodied experience. Drawing on interdisciplinary work across HCI, immersive media, affective computing, and critical theory, we argue that bioadaptive media constitute a form of techno-biological infrastructure that raises epistemic, aesthetic, and ethical questions. We present a preliminary work-in-progress sociotechnical taxonomy of such systems and offer a framework for assessing their implications for user experience, agency, and normative assumptions about cognition, affect, and perception.

CCS Concepts

• Human-centered computing → HCI theory, concepts and models; HCI design and evaluation methods.

Keywords

Affective Computing, Perceptive Systems, User Modelling, Individuation, Experiential Evaluation, Critical HCI

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1 Introduction

Bioadaptive media, as referred to in this paper, stands for systems that not only adapt visual, spatial, or narrative content based on users' real-time physiological states, but also recursively modulate those states through continuous perceptual feedback. These systems are emerging across XR, therapeutic applications, games, and adaptive learning, modulating perception based on psychophysiological signals such as arousal, attention, or emotional valence [5, 12, 15], and recently aided by the advances in biosensing. This field lies at the intersection of established traditions, such as affective computing [9, 17], seeking to sense or model emotional states

through physiological data; user modelling and personalisation (e.g., [13]), inferring preferences or user profiles; and algorithmic adaptation [14, 20], as in recommender systems.

While these traditions have enabled the technical development of adaptive systems, they also carry embedded assumptions such as the legibility of emotion from biosignals, the desirability of optimisation, and the idea of the user as a regulatable input-output system. These assumptions raise known issues in HCI and AI ethics in a novel and arguably more intimate context, including reductionism, data bias, affective opacity, and performative misrepresentation [1, 2]. As these paradigms are extended and recombined into immersive, real-time, perceptually recursive environments, they also create novel assumptions. Recursive bioadaptive systems directly model the perceptual conditions of experience, creating elaborate perceptual infrastructures [11]. They move from representational inference and symbolic interaction to perceptual co-regulation, altering the user's sensorial experience through adaptable content in direct and real-time response to internal states. This raises critical design and evaluative questions that require urgent transdisciplinary attention: How are emotions and attention operationalised? What norms about perception and cognition are encoded into adaptive responses? What forms of agency do users retain when their environments adapt implicitly and continuously to biometric cues? We address these questions through a preliminary conceptual framework and taxonomy in support of critical analysis of bioadaptive media as perceptual infrastructures.

2 Recursive Perceptual Modulation as Techno-Biological Infrastructure

Work on augmented human interaction often emphasises sensory extension through enhancing range, resolution, or modality of perception [16]. Recursive bioadaptive systems differ as they do not extend sensation, but instead regulate the conditions of perception itself via the same or similar sensorial modality. For example, an virtual environment that modifies pacing or colour saturation in response to HRV is not adding a new sense but altering users' perception of the environment and affective experience nonetheless.

This type of media technology implicates the user in a continuous loop of mutual adaptation that could often be opaque and non-negotiable, turning users into objects of the modulation rather than agentic subjects and users of a system. Understanding the degrees and modalities of agency requires distinguishing among the different forms it can take. We can recognise at least three potential scenarios, all of which can be imagined in an array of application domains: *interpretive agency*, where users can understand or become aware of how the system is modulating their environment; *configurative agency*, where users have tools to adjust or override



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adaptation parameters; and *subversive agency*, where users can opt out, subvert, or contest the system's assumptions or feedback logics. These forms of agency map onto design choices such as perceptibility, feedback logic, and the availability of user controls. For example, systems that provide open-loop control or explicit (bio)feedback afford higher interpretive and configurative agency, whereas closed-loop, opaque systems tend to limit all three forms. Understanding these mappings is essential for designing sustainable systems that employ recursive bioadaptation.

Moreover, drawing on Simondon's theory of individuation [18], we interpret bioadaptive media as infrastructures that co-constitute perception through techno-biological feedback. Individuation emphasises that one emerges through dynamic relations with structures, whereas these bioadaptive systems arguably form such significant and deeply problematic structures. This theoretical lens is valuable as it shifts attention from the user as an isolated, autonomous, agentic individual to the user as an individual emergent through relational coupling with adaptive technical systems. Rather than treating recursive bioadaptive interfaces as neutral tools that represent a simple extension of existing paradigms, we emphasise individuation as grounds to critique how bioadaptive systems may foreclose perceptual variability or reinforce normative affective states under the guise of personalisation or optimisation. As it has been suggested before [10], real-time computational systems can already increasingly shape what is perceived before conscious awareness, while by intentionally structuring attention, pacing, and visual dynamics in response to biometric data, these systems could implicitly further reinforce dominant cognitive or affective norms.

3 A Taxonomy of Perceptual Infrastructures

To support responsible, inclusive, and sustainable design as well as analysis and comparative research, we propose a preliminary six-dimensional taxonomy building on and expanding on existing design-oriented work [15] that outlines four key components, namely Medium, Application Area, Psychophysiological Target State, and Adaptation Technique (MAPA framework). Such existing taxonomies and frameworks in relation to recursive bioadaptive media focus solely on technical and design issues, but nevertheless present a valuable basis for understanding and critically reflecting on various components and affordances of these systems. For example, while MAPA's Psychophysiological Target State focuses on identifiable affective or cognitive states (e.g., attention, arousal), our category of Perceptual Target broadens this to include how systems aim to modulate perception itself through rhythm, salience, or structure. Similarly, MAPA's Adaptation Technique is expanded into two distinct but interrelated categories of Feedback Logic, which addresses the temporal and operational structure of adaptation (e.g., closed loop vs. semi-closed loop), and Modulation Channel, which can open up how adaptation is materially enacted. Furthermore, our taxonomy introduces categories that MAPA omits entirely but that are crucial for understanding the user's situated experience and the normative assumptions embedded in system design. User Perceptibility addresses the extent to which modulation is perceivable, interpretable, or opaque to the user; System Objective, as extended

from MAPA's Application Area, also allows for speculative, aesthetic, or critical design intents that exceed conventional use cases such as therapy or training.

Finally, we recognise as relevant the following: (1) what is being sensed, for what (2) affective/perceptual target and (3) system objective, (4) how the input is interpreted and structured through a particular feedback logic, (5) what parameters of the media are modulated, and finally (6) whether and how the user perceives, interprets, or resists this modulation.

- (1) *Input Modality*: the physiological or behavioural data sensed (e.g., EEG, HRV, GSR, eye tracking).
- (2) *Perceptual Target*: the experiential domain modulated (e.g., affect, attention, perceptual flow).
- (3) *System Objective*: what system aims at achieving (e.g., therapeutic regulation, exploratory play, persuasive influence).
- (4) *Feedback Logic*: the structure of system adaptation—closed loop (continuous automatic modulation), semi-closed loop (discrete, threshold-based responses), or open loop (manual or externally triggered changes).
- (5) *Modulation Mode*: what is modulated (e.g., color, rhythm, spatial structure, narrative pace).
- (6) *User Perceptibility*: how visible or felt the adaptation is; explicit (user is informed or can intervene), implicit (hidden but experienced), or performative (user engages actively with system).

We could shortly consider two plausible but distinct speculative systems as examples illustrating different configurations: a) a biofeedback-driven XR mindfulness application designed for workplace productivity, and b) an artistic installation that modulates audiovisual patterns in response to collective electrodermal response levels. The former typically features a closed-loop logic with low perceptibility and a therapeutic objective, reinforcing a normative model of calmness and focus. In contrast, the latter may employ a semi-open or performative feedback model, making the modulation explicit and inviting interpretive or even subversive forms of user agency.

4 Methodological Implications and User Experience Assessment

Traditional HCI evaluation commonly focuses on task performance, usability, and self-reported aspects of interaction, such as satisfaction with the system, which largely disregards the recursive, embodied, and temporal nature of recursive bioadaptive experiences. While we advocate for alternative methods that address these gaps, more specificity is needed regarding their deployment, analytical frameworks, and validity. For example, physiological-trace ethnography (for trace ethnography, see [7]) must be more than just a juxtaposition of sensor logs and interviews but rather account for interpretation bias, affective ambiguity, and the non-obvious mapping between physiology and felt experience. Similarly, phenomenological walkthroughs [8] risk superficiality unless grounded in a rigorous interpretive framework. Anticipatory and speculative design methods ([4, 19], while valuable for surfacing normative tensions, can risk abstraction or detachment from existing systems if not grounded in empirical or participatory insight. Finally, critical scenario testing [3, 6], while powerful for surfacing design

assumptions, may lack systematic procedures for comparative evaluation. Despite these limitations, we outlined these four angles as promising for future development. The taxonomy, in turn, supports such work by offering a vocabulary for articulating what kinds of adaptation are occurring and how users may or may not perceive, interpret, or influence them. Nonetheless, further research should refine how these methods can be validated, standardised, and scaled in the domain of recursive bioadaptive media, especially for evaluating systems with experiential effects unfolding dynamically and often implicitly. In summary:

Physiological-trace ethnography. Combining biosensing logs with situated user interviews or field observation. It can reveal discrepancies between system-detected states and user experience, emphasising the potential interpretive mismatch between computation and experience and can uncover how users negotiate, ignore, or perform responses to bioadaptive systems. For instance, it can reveal mismatches in perceptual target (e.g., whether arousal truly maps onto attention) or system objective (e.g., calming vs. alerting).

Phenomenological walkthroughs. Guiding participants through a stepwise reenactment or real-time use of a bioadaptive system, while prompting them to describe their evolving perceptual and affective states. It is particularly suited to uncovering how modulation is felt over time, what aspects are perceptible, and how users make sense of their changing experiential context. This aligns closely with evaluating user perceptibility, modulation mode, and user agency, especially interpretive and resistive dimensions.

Anticipatory futures and speculative design research. Extending beyond analysis of existing or mock-up designs, these approaches aim to explore the social, cultural, and ethical implications of recursive bioadaptive systems before they are fully realised. Through scenario development, design fiction, or counterfactual prototyping, anticipatory methods can help surface latent tensions in how future perceptual infrastructures may distribute agency, encode affective norms, or shape epistemic environments.

Critical scenario testing. Complementary to anticipatory and speculative methods, this includes presenting users with exaggerated, adversarial, or edge-case adaptations (e.g., a system that misclassifies relaxation as stress) to probe the robustness and values embedded in feedback logic. It can also involve speculative scenarios that prompt reflection on plausible futures.

5 Conclusion

Recursive bioadaptive media represent a conceptual and practical shift in how digital systems interact with users. These systems deepen and expand issues of agency, transparency, literacy, and normativity. By reframing them as perceptual infrastructures and offering a critical taxonomy, we aim to enable deeper inquiry into how such systems encode, reinforce, or resist assumptions around perceptual modulation.

Future research should not only build systems that are technically robust but also reflectively engaged with the cultural, ethical, and experiential stakes of perception-oriented adaptation, as well as anticipatory or emerging and future developments of related systems. This includes considering the diverse cognitive and affective needs of users, supporting user agency, and enabling interpretability of feedback processes, as well as wider sociocultural

implications, potential societal disruptions, and user literacy. The proposed taxonomy could thus assist designers during the ideation and prototyping phases by helping them identify the perceptual targets, feedback structures, and visibility levels, while providing a vocabulary for articulating trade-offs between adaptation and agency, enabling critical decisions about when and how adaptation should be implemented, disclosed, or made configurable. Similarly, it could aid anticipatory and critical research by providing a basis for interrogation of recursive bioadaptive systems.

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